

ISYS1055/1057/3412 (Practical) Database Concepts

Assessment 2: SQL Programming and Normalization

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/*

Part B

Problem 1

*/

SELECT b.title, b.subtitle, b.edition, b.place

FROM book b

INNER JOIN published_by pb

ON b.bookdescid = pb.bookdescid

WHERE LOWER(b.place) LIKE '%oxford%'

AND LOWER(role) LIKE 'publisher';

/*

Problem 2

*/

SELECT b.bookdescid, COUNT(bc.bookdescid) AS "number of copies"

FROM book b

INNER JOIN book_copy bc

ON b.bookdescid = bc.bookdescid

GROUP BY bc.bookdescid;

/*

Problem 3_A

*/

```
SELECT DISTINCT b.bookdescid, b.title, b.subtitle, b.edition, b.place
FROM book b
LEFT JOIN borrow_copy brc
ON b.bookdescid = brc.bookid
WHERE brc.transactionID IS NULL;
```

/*

Problem 3_B

*/

```
SELECT DISTINCT b.bookdescid, b.title, b.subtitle, b.edition, b.place
FROM book b
WHERE b.bookdescID NOT IN (
SELECT DISTINCT bc.bookID
FROM borrow_copy bc
);
```

/*

Problem 4

*/

```
SELECT a.firstname, a.middlename, a.lastname, COUNT(w.bookdescid) AS 'booknumb'
FROM author a
INNER JOIN written_by w
ON a.authorid = w.authorid
WHERE w.role = 'Author'
GROUP BY a.authorid
HAVING booknumb >1 ;
```

/*

Problem 5

*/

```
SELECT b.title, DATE(br.returndate) AS 'returnDate',  
DATE(br.duedate) AS 'dueDate',  
julianday(br.returndate) - julianday(br.duedate) AS 'daysDelay'  
FROM borrow br  
JOIN borrow_copy brc  
ON br.transactionid = brc.transactionid  
JOIN book b  
ON brc.bookid = b.bookdescid  
WHERE julianday(br.returndate) > julianday(br.duedate);
```

/*

Problem 6

*/

```
SELECT p.publisherfullname  
FROM publisher p  
LEFT JOIN published_by pb  
ON p.publisherID = pb.publisherID  
AND pb.role = 'Editor'  
WHERE pb.publisherID IS NULL  
ORDER BY p.publisherfullname;
```

/*

Problem 7

*/

```
SELECT a.firstname || ' ' || a.lastname AS 'author',
```

```

b.title AS 'bookTitle',
s.subjecttype AS 'bookSubject',
b.year AS 'publishedYear'
FROM author a
INNER JOIN written_by wb
ON a.authorID = wb.authorID
INNER JOIN book b
ON wb.bookdescID = b.bookdescID
INNER JOIN subject s
ON b.subjectID = s.subjectID
WHERE s.subjecttype = 'Artificial Intelligence';

```

/*

Problem 8

*/

```

SELECT b.title AS 'bookTitle',
p.firstname || ' ' || p.lastname AS 'borrowerName',
DATE(br.borrowdate) AS Date_of_Borrow
FROM book b
JOIN borrow_copy brc
ON b.bookdescid = brc.bookid
JOIN borrow br
ON brc.transactionid = br.transactionid
JOIN person p
ON br.personid = p.personid
WHERE p.personid IN (SELECT p.personid
FROM book b
JOIN borrow_copy brc
ON b.bookdescid = brc.bookid

```

```
JOIN borrow br
ON br.transactionid = br.transactionid
JOIN person p
ON br.personid = p.personid
WHERE LOWER(b.title) LIKE 'computer science')
AND LOWER(b.title) != 'computer science';
```

```
/*
```

Problem 9

```
*/
```

```
SELECT b.bookdescid, b.title, b.year,
COUNT(br.transactionid) AS 'timesBorrowed'
FROM book b
LEFT JOIN borrow_copy br
ON b.bookdescid = br.bookid
GROUP BY b.bookdescid
HAVING COUNT(br.transactionid) > 0;
```